BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 025 · Network Topologies

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does DNS do?

Q2. Why is a thermostat controller an embedded system?
A. It has a specific task
B. It runs any desktop app
C. It always needs Wi-Fi
D. It always has a screen

Q3. What makes a system embedded?

Q4. A circuit uses two voltage ranges for two states. “Computers use binary because people cannot write numbers above 1.” Correct this. Why are two states useful?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

I Do · Read new knowledge

Key vocabulary

Topology — Arrangement of network devices and links.

Star topology — Endpoints connect to a central switch or hub.

Mesh topology — Devices connect via multiple paths between nodes.

Fault tolerance — Ability to continue working after some faults.

Central switch — Network device that connects and forwards data to LAN devices.

Redundancy — Extra paths that can be used if a path fails.

New learning

Topology describes connections: A network topology shows how computers and connection devices link together.

Star topology: Each endpoint connects to a central device such as a switch. Easier to add/manage devices, but the central device is a point of failure.

Mesh topology: Devices can connect to multiple others. Alternative paths make the system resilient if one link fails, but more connections increase cost and complexity.

Compare in context: For a small school LAN, a star can be practical. A high-availability network may use mesh routes; explain what happens when a link or central device fails.

Worked example

A school LAN uses a star topology. Explain one advantage and one disadvantage if its central switch fails. [4 marks]

1. All classroom devices connect to one central switch.

2. New devices can be connected easily [1], making the network straightforward to manage [1].

3. The switch is a single point of failure [1], so its failure can interrupt communication for connected devices [1].

4. Give advantage plus linked benefit and disadvantage plus linked effect.

We Do · Practise with guidance

Guided 1

A business needs reliable connections between several sites. Explain one advantage and one disadvantage of a mesh topology. [4 marks]

Q3. Star devices connect to a central switch [1], are easier/cheaper to manage [1], but depend on the central device [1]. Mesh provides more than one path [1], so it can withstand some link failures [1], but uses extra connections increasing cost/complexity [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Distributed DNS servers help look up the IP address associated with a domain name.

Q2. A — It has a specific task

Q3. It is built into a larger device for a specific task, often with limited resources.

Q4. Circuits can reliably tell two states apart: 0 and 1. These states represent all kinds of data in binary.

I Do / We Do pages · Guided practice

Q1. Multiple possible routes [1] mean traffic can use another path if a link fails [1]. More connections/hardware are needed [1], increasing cost or maintenance [1].

Q2. Each device has its own link to the central device [1], making fault isolation easier [1]. Devices can be added or managed centrally [1], simplifying expansion [1].

Q3. Mesh provides multiple paths between devices [1]. When a link fails [1], traffic may be routed along another available path [1], keeping communication possible [1].

You Do pages · Code First

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

Expected result: 6

You Do pages · Independent questions

Q1. Several routes provide fault tolerance if a link fails [1].

Q2. The central switch connects the kiosks/computers [1], so its failure can interrupt network communication [1]. A spare or quickly replaceable switch, or a resilient design, can reduce the problem [1] by restoring service/reducing downtime [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: Multiple possible routes [1] mean traffic can use another path if a link fails [1]. More connections/hardware are needed [1], increasing cost or maintenance [1].

Guided 2

A small primary school chooses a star topology. Explain two advantages of this decision. [4 marks]

Worked method: Each device has its own link to the central device [1], making fault isolation easier [1]. Devices can be added or managed centrally [1], simplifying expansion [1].

Guided 3

Describe how a mesh can continue operating after one connection fails. [4 marks]

Worked method: Mesh provides multiple paths between devices [1]. When a link fails [1], traffic may be routed along another available path [1], keeping communication possible [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the count of links for a small full mesh.

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

predict

Question 1

State one advantage of a mesh topology. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.1.

Question 2

A cinema uses a star network to connect ticket kiosks and staff computers. The central switch stops working. Explain the effect and suggest one precaution. [4 marks] Source: Original J277-style question, not an OCR past-paper original.

Question 3

Compare star and mesh topologies for a large organisation. [6 marks] Source: Original J277-style question, not an OCR past-paper original.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Mesh topology generally provides alternative links that can improve fault tolerance.